

7 complete genomes (PRJNA357031), RefSeq PGAP annotation

33,170 CDS · 9 chromosomes, 2 plasmids

A · GFF text parsing (gene symbol / product)

unique name/product: 1,979
named once in all 7: 1,022

B · Protein homology (mmseqs2 all-vs-all)

no homolog in other 6: 7,411
1:1 in all 7 genomes: 979

C · DNA uniqueness (18-mers, jellyfish)

unique fraction of primary
chromosome: 65–92 %

Eligible candidate genes

6,939 genes · either route, chromosomal, not pseudo, ≥ 300 bp, > 5 kb from mobile elements

Sequence-aware filter: nucleotide uniqueness + single copy

5,087 genes on the dnaA chromosome · ≥ 150 bp unique DNA, single BLAST hit in own genome

Primer / probe design (primer3)

1,078 assays on 552 genes · 40 core + 40 unique genes per strain, 70–150 bp amplicons

Exhaustive in-silico PCR (bowtie1, ≤ 3 mismatches, 7 genomes + plasmids)

716 assays pass · single on-target product, no near-perfect off-target primer site

Multiplex selection

42 assays (top 6 genes per strain) → 279,936 7-way combinations scored on heterodimer ΔG

Final 7-plex, verified in silico

pooled in-silico PCR: exactly 7 products · knock-out 7/7 · worst between-assay ΔG -5.64 kcal/mol